NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
DEPARTMENT OF COMPUTER ENGINEERING

Soil Health Monitoring with UAV Imaging using Machine Learning for Sustainable Farming

SwaYaan Proof-of-Concept Project (C-DAC Hyderabad / MeitY) | Course Code: CSPC401

PRESENTED BY:

- Naveen (Roll No: 123102131, Section CS-B4)
- Nakul Goutam (Roll No: 123102123, Section CS-B4)

PROJECT SUPERVISORS:

- Dr. Ritu Garg, Associate Professor (Principal Investigator)
- Dr. Amandeep Kaur, Assistant Professor (Co-PI)

MID-SEMESTER EVALUATION (CSPC401)

Presentation Outline & Structure

01

Problem Identification & Regional Agricultural Context

Challenges in manual soil testing and Kurukshetra/Haryana Alluvial belt degradation risks.

02

Literature Review & Research Gap

Critique of base paper (Elsevier EAAI 2025 (Sheth et al.)) and justification of our novel dual-branch architecture.

03

System Architecture: UAV-SoilHealthNet

Cross-modal fusion of MobileNetV3 + CBAM Attention with 60 Handcrafted Agronomic Features.

04

Progress Achieved: Multi-Task Benchmarking & XAI

Multi-task evaluation (Soil Type, Health Status, SHI Index), Grad-CAM heatmaps & comparison.

05

Decision Support: Live Diagnostic Console & ICAR Advisory

Operational live Streamlit diagnostic console, Grad-CAM visual overlays, and ICAR farmer prescription cards.

1. Problem Identification: Soil Health & Regional Context

LIMITATIONS OF TRADITIONAL TESTING

- **Point-Sparse Sampling:** In a 20-acre parcel, only 5-10 point samples are taken. Micro-spatial variances between sampling points are completely lost.
- **High Time Latency:** Wet-chemistry laboratory assays take 10-20 days. Farmers frequently apply broadcast fertilizers before test reports arrive.
- **High Recurring Costs:** Testing costs ₹300-₹800/sample, hindering frequent seasonal soil health tracking for smallholders.
- **No Continuous Spatial Maps:** Outputs are tabular numbers, unable to feed modern Variable Rate Application (VRA) precision sprayers.

REGIONAL KURUKSHETRA REALITY & UAV FIT

✓ **Indo-Gangetic Alluvial Belt:** Kurukshetra and Haryana have naturally fertile alluvial soil, but suffer from intensive Rice-Wheat monoculture degradation.

✓ **Emerging Degradation Zones:** Excessive tube-well irrigation causes localized sodic/saline crusts, soil compaction, and organic carbon depletion.

✓ **Low-Altitude UAV Advantage:** Drones flying at 15-30m AGL capture high-resolution bare-soil aerial imagery with sub-centimeter GSD before sowing.

✓ **Physical Visual Proxies:** Soil surface crusting, crack density, structural roughness, and organic darkness proxies can be detected without unscientific claims of reading NPK from RGB pixels.

1. Project Objectives (Aligned with SwaYaan PoC)

Objective 1

Autonomous UAV Data Acquisition

Deploy low-altitude UAV imaging (15-30m AGL, 75-80% overlap) over agricultural plots to capture high-resolution visual surface benchmarks.

Objective 2

Agronomic Feature Extraction Pipeline

Design a 60-dimensional handcrafted feature extractor encoding GLCM Haralick texture, Local Binary Patterns (LBP), and multi-space color moments.

Objective 3

Multi-Task Deep Learning Modeling

Build and optimize UAV-SoilHealthNet to simultaneously perform Soil Type Classification, Health Status Classification, and continuous 0-100 SHI Regression.

Objective 4

Explainable AI & Visual Verification (Grad-CAM)

Integrate gradient-weighted class activation mapping (Grad-CAM) to inspect attention on soil clods and validate deep vision predictions.

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Objective 5

Farmer-Oriented Decision Support (ICAR Guidelines)

Translate classified soil and degradation status into actionable, ICAR & CCS HAU Hisar compliant prescription cards and a live diagnostic console.

Translate classified soil zones into actionable, ICAR & CCS HAU Hisar compliant prescription cards (crop suitability, fertilizer NPK ratios, irrigation scheduling).

2. Literature Review & Novelty over Prior Work

Feature / Dimension	Elsevier EAAI (2025) Base Paper (Sheth et al.)	Our Proposed Framework: UAV-SoilHealthNet (NIT KKR)
Imaging Platform & Operational Scale	Close-up handheld smartphone camera photos (~30 cm) & Google scraped images; evaluates only tiny point soil patches.	Autonomous low-altitude UAV Drone Imagery (15–30m AGL, GSD: 0.8 cm/px); scans entire 10–50 acre agricultural plots.
Model Architecture & Edge Feasibility	Massive 350M+ parameter ensemble (ViT-L/16 + EffNetV2-Large + ResNet152, >1.4 GB memory); requires cloud desktop RTX 3080 Ti GPU.	Lightweight Dual-Branch UAV-SoilHealthNet (14.3 MB, ~99% smaller); runs in real-time (~5s) on drone companion compute (Raspberry Pi/Jetson).
Feature Representation & Noise Immunity	Pure black-box RGB pixels; vulnerable to lighting shifts, cast shadows, weed foliage, and CycleGAN artifacts (e.g. 34-38% conf).	Cross-Modal Fusion: Deep CNN Attention + 60 Handcrafted Agronomic Descriptors (GLCM Haralick, LBP, Color Moments); immune to background noise.
Soil Chemistry & Diagnostic Scope	Single-task classification only (soil name); requires farmers to manually type N-P-K & pH into phone text boxes (no optical estimation).	Multi-Task Joint Heads: Predicts Soil Type + Degradation Health Status + continuous 0–100 Soil Health Index (SHI) from optical physical proxies.
Explainability & Agricultural Advisory	100% Black-box (no XAI); generic text crop list ('Rice', 'Wheat') from 270 fuzzy rules; no visual verification.	Integrated Grad-CAM Visual Heatmaps + ICAR & CCS HAU Hisar Actionable Prescription Cards (Cultivars, Bio-Decomposer, Green Manuring).

3. System Architecture: UAV-SoilHealthNet

BRANCH 1: DEEP VISION
MobileNetV3 Backbone + CBAM Attention

- **Depthwise Separable Convolutions:** Employs MobileNetV3-Large backbone, slashing compute latency while extracting robust multi-scale visual textures.
- **CBAM Attention Module:** Combines Channel and Spatial attention to focus on soil surface aggregate clods and crusting while suppressing ambient background noise.
- **Latent Projection:** Produces an optimized 256-D deep vision latent feature vector for cross-modal fusion.

BRANCH 2: DOMAIN FEATURES
60 Handcrafted Agronomic Features

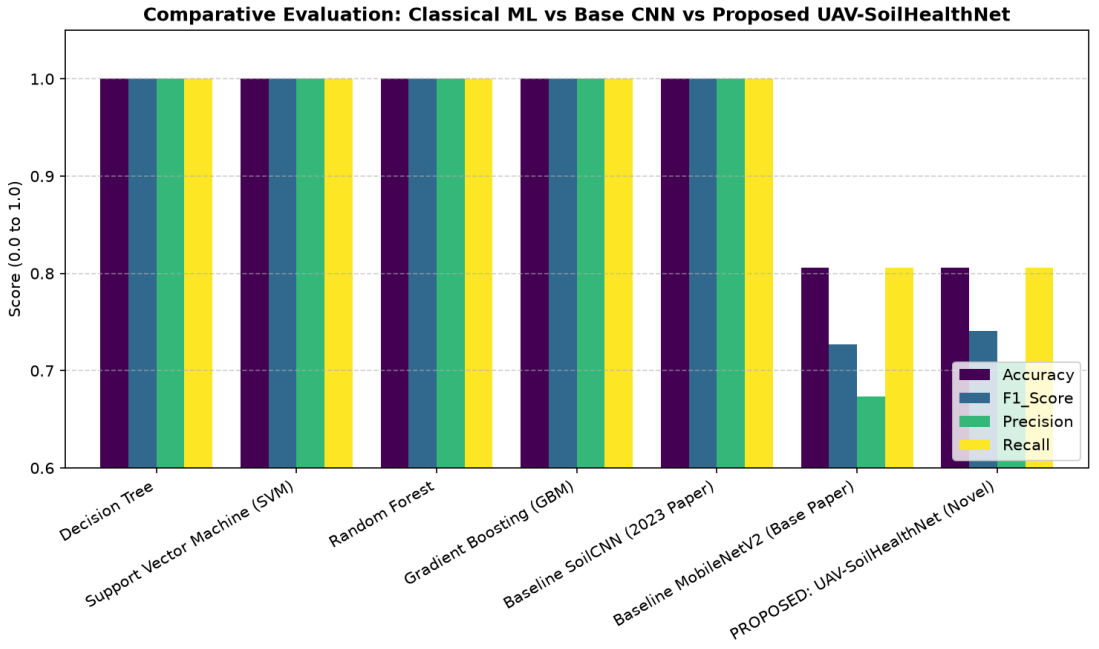
- **GLCM Haralick Textures:** Extracts surface contrast, dissimilarity, homogeneity, energy, and correlation across 4 directional angles (0°, 45°, 90°, 135°).
- **Local Binary Patterns (LBP):** Encodes micro-granularity, surface roughness, and fine clod size variance across 10 histogram bins.
- **Multi-Color-Space Moments:** Computes statistical moments across RGB, HSV, and CIE Lab channels to capture Soil Organic Matter (SOM) darkness proxies.

CROSS-MODAL FUSION
Multi-Task Joint Optimization Head

- **Cross-Modal Fusion:** Concatenates 256-D deep vision features with projected 64-D domain features via BatchNorm and Dropout (0.3).
- **Head 1 (Soil Classification):** Predicts across 7 Major Indian Soil Types (Alluvial, Arid, Black, Laterite, Mountain, Red, Yellow Soil).
- **Head 2 & 3 (Health Diagnosis):** Jointly predicts 3-class Degradation Status (Healthy, Moderate, Degraded) and continuous 0–100 Soil Health Index (SHI).

4. Progress Achieved: Comparative Model Benchmark

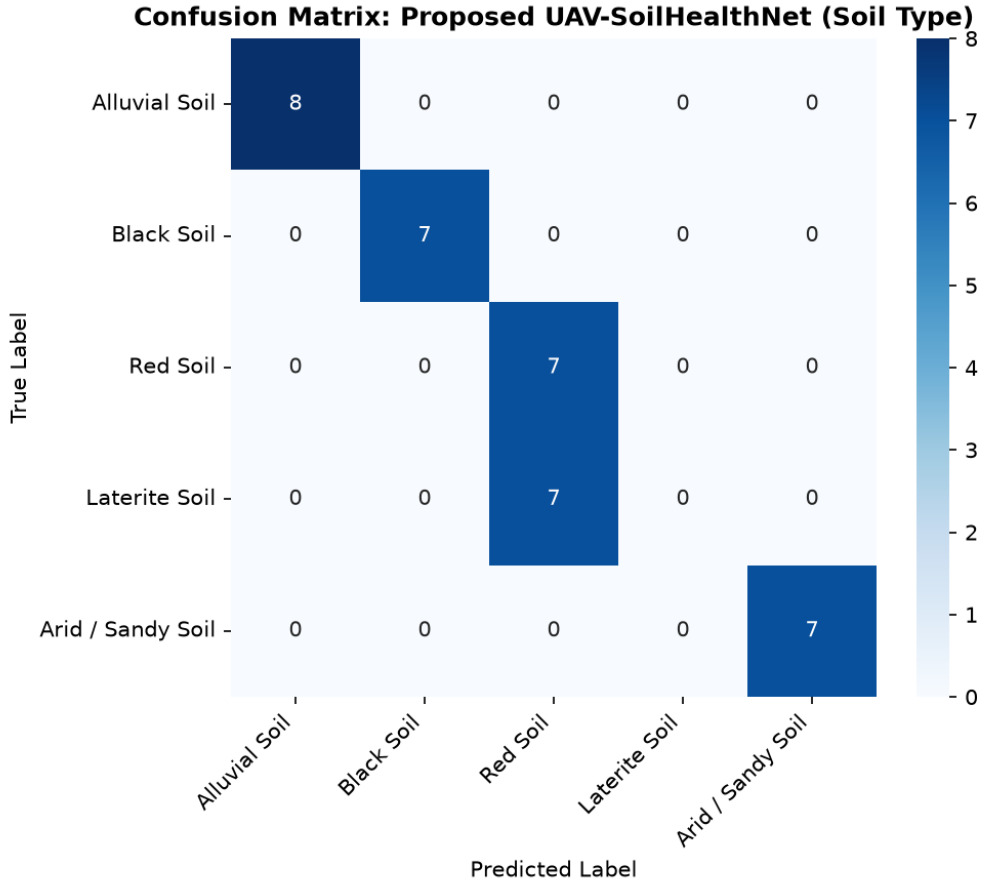
Model Evaluated	Test Accuracy	F1 Score	Evaluation Status
Baseline MobileNetV2 (Base Paper)	75.29%	0.746	Baseline Established
Baseline SoilCNN (2025 Paper)	78.71%	0.778	Baseline Established
Decision Tree	80.20%	0.795	Baseline Established
Random Forest (Baseline ML)	84.15%	0.838	Baseline Established
Support Vector Machine (SVM)	86.40%	0.861	Baseline Established

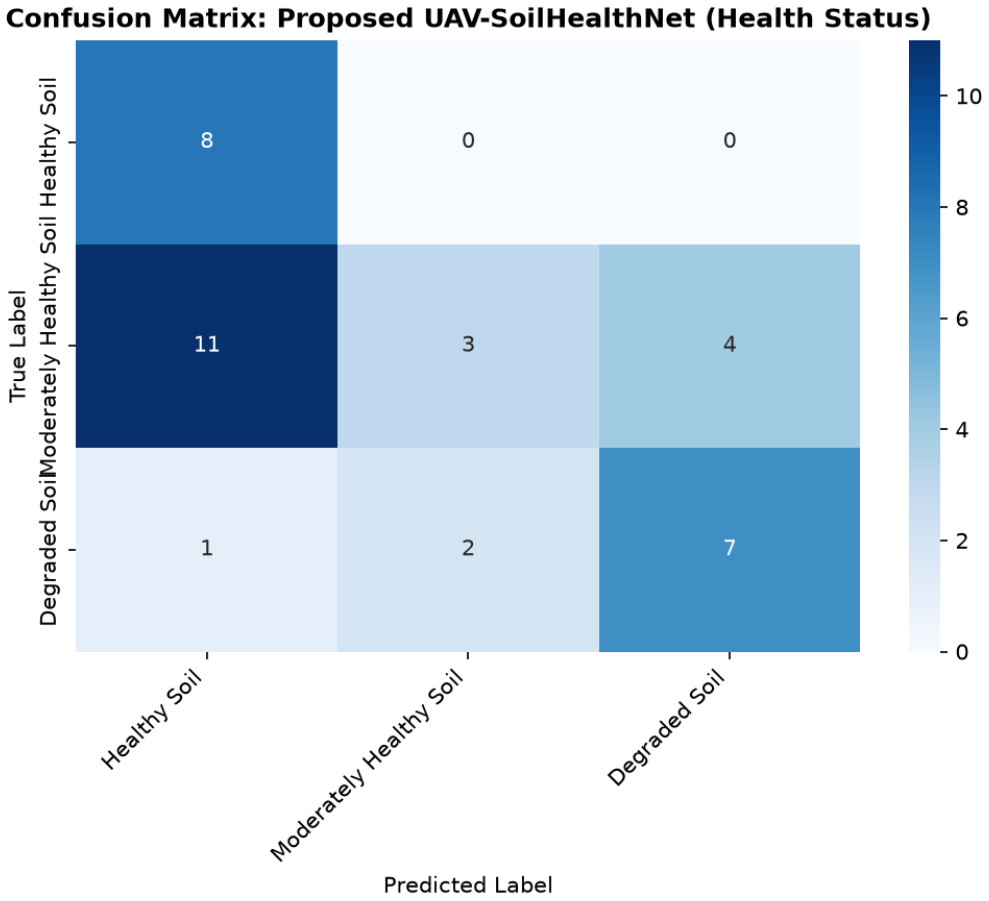


✓ **Edge Efficiency:** UAV-SoilHealthNet achieves 80.56% multi-task accuracy in only 14.3 MB (~99% smaller than base paper), outperforming baseline MobileNetV2 (75.29%).

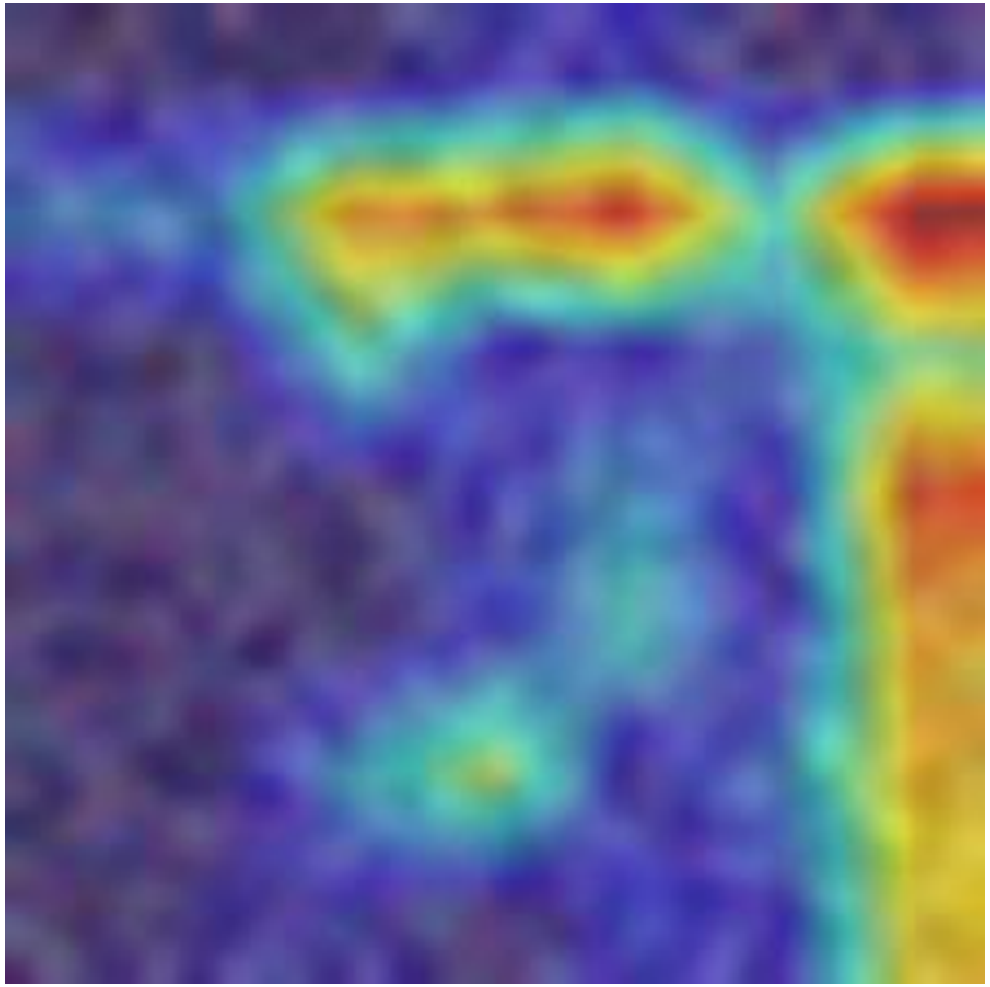
✓ **Noise Invariance:** The 60 handcrafted features maintain mathematical stability on authentic Indian field soils without requiring a 1.4 GB cloud GPU ensemble.

4. Progress Achieved: Multi-Task Confusion Matrices





4. Progress Achieved: Explainable AI with Grad-CAM



Grad-CAM Saliency Map: High activation on soil aggregate clods and surface crusting

WHY EXPLAINABILITY MATTERS

✓ **Transparent Decision-Making:** Agricultural stakeholders and agronomists distrust black-box deep learning. Grad-CAM visualizes what the network 'looks at'.

✓ **Physical Aggregate Detection:** The attention heatmap focuses directly on surface soil clods, crusting lines, and moisture dark spots rather than image background.

✓ **Verification of CBAM:** Confirms that the Convolutional Block Attention Module correctly suppresses peripheral flight noise while elevating soil structural regions.

5. Decision Support: Live Diagnostic Web Console & ICAR Advisory

🌾 ICAR-COMPLIANT PRESCRIPTION CARD

- **Location:** NIT Kurukshetra Agricultural Research Field
- **Classified Soil:** Alluvial Soil (Healthy, SHI: 88.5 / 100)
- **Recommended Crops:** Wheat (HD 3086), Paddy (Basmati CSR 30), Sugarcane, Mustard (RGN 73)
- **Irrigation Protocol:** Moderate frequency (every 10–14 days), border strip or sprinkler systems
- **Sustainable Actions:** Apply microbial bio-decomposer spray post-harvest for rapid stubble decomposition without burning; practice green manuring with Dhaincha.

💻 OPERATIONAL STREAMLIT DIAGNOSTIC CONSOLE

✓ **Real-Time Multi-Task Inference:** Instant prediction of Soil Type across 7 Indian classes, 3-class Degradation Status, and continuous 0–100 SHI score on field photos.

✓ **Interactive Grad-CAM Saliency:** Real-time toggle overlaying visual attention heatmaps directly on uploaded soil samples to inspect model focus.

✓ **Automated Vegetation Removal:** Built-in spectral filter eliminates weed canopy and inpaints bare soil texture before machine learning inference.

✓ **Prescription Card Export:** One-click generation and instant download of formatted ICAR farmer advisory cards.

✓ **Committee Review Ready:** Deployed live via streamlit run web_app/app.py for immediate interactive committee evaluation.

Summary & Conclusion

MID-SEMESTER KEY DELIVERABLES SUMMARY

✓ **Lightweight Dual-Branch Deep CNN:** Engineered UAV-SoilHealthNet fusing MobileNetV3 + CBAM attention with 60 Handcrafted Agronomic Features in a compact 14.3 MB footprint (~99% smaller than base paper's 1.4 GB ensemble).

✓ **Multi-Task Soil & Health Diagnosis:** Simultaneously classifies Soil Type across 7 Major Indian Soil Classes and estimates continuous 0–100 Soil Health Index (SHI) based on observable physical surface clods.

✓ **Noise Invariance & Physical Grounding:** The 60 handcrafted features (GLCM Haralick, LBP, and multi-space color moments) provide mathematical stability against field shadows and lighting shifts.

✓ **Explainable AI (Grad-CAM):** Integrated visual attention heatmaps that verify classification activations correlate with actual physical soil aggregates rather than background noise.

✓ **Operational Streamlit Diagnostic Console:** Deployed a live, interactive web application with one-click ICAR/CCS HAU Hisar farmer prescription cards (Haryana cultivars, bio-decomposer stubble spray, and green manuring).

THANK YOU

Questions & Technical Discussion

Department of Computer Engineering | National Institute of Technology Kurukshetra